

Supporting Information for
A sharp melt-bearing lithosphere–asthenosphere boundary beneath the
Korean Peninsula: Evidence from teleseismic receiver functions

Minkyung Kim^{1,2}, Hyunsun Kang^{1,2}, Gyuchan Park^{1,2}, Jung-Woo Park^{1,2}, and YoungHee Kim^{1,2†}

¹Research Center for Deep-Surface Coupling of Earth, Seoul, Republic of Korea

²School of Earth and Environmental Sciences, Seoul National University, Seoul, Republic of Korea

Corresponding author: YoungHee Kim (younghkim@snu.ac.kr)

Contents of this file

Texts S1 to S5

Figures S1 to S12

Introduction

This supplementary material consists of four main components. First, we describe the P-receiver function processing procedures, including the quality-control steps and the corrected $H-k-c$ stacking procedure, together with curve-fitting results used to validate the inferred Moho depth and crustal P-to-S wave velocity ratio (V_P/V_S). Second, we present thermodynamic modeling results assessing the sensitivity of melt-fraction estimates to lithospheric composition and lithosphere–asthenosphere transition schemes. Third, we provide a consistency test comparing Moho depths independently estimated from P- and S-receiver functions. Finally, we use Monte Carlo uncertainty propagation to evaluate whether the observed regional differences in Moho depth, crustal V_P/V_S , and LAB depth remain robust within the estimated station-specific uncertainties.

Text S1. Quality control after computing PRFs

Quality control for the PRFs was applied based on the following criteria. First, the direct P arrival was required to show positive polarity. Second, the P -to- S converted phase from the Moho (Pms) was required to arrive 3–5 s after the direct P arrival, consistent with the expected crustal thickness beneath the KP (e.g., Chang & Baag, 2007). Stations on Ulleung Island were excluded from this criterion because their unusually thin crust results in earlier Pms arrivals; PRFs from these stations were examined manually. Third, the maximum amplitude of the tangential RF was required not to exceed that of the radial RF. After quality control, 75,413 PRFs were retained for further analysis.

Text S2. Corrected H - κ stacking

At several stations, conventional H - κ stacking was inadequate because the RFs exhibited strong signatures of dipping interfaces, crustal anisotropy, and/or sediment reverberations (Li et al., 2019; Yu et al., 2015). To account for these effects, we applied the extended H - κ - c stacking scheme of Li et al. (2019) to stations on Jeju Island, Ulleung Island, and the Jiri Mountain area (Figure S2), where such complexities are particularly pronounced.

For stations affected by sedimentary reverberations, we first applied the resonance-removal procedure of Yu et al. (2015). For each RF, the normalized autocorrelation function was used to determine the reverberation strength (r_0) and two-way traveltime (Δt) from the amplitude and time lag of the first negative peak, respectively. The sediment reverberations were then suppressed in the frequency domain using the resonance-removal filter

$$F(i\omega) = H(i\omega)(1 + r_0 e^{-i\omega\Delta t}) \quad (1)$$

where $H(i\omega)$ and $F(i\omega)$ denote the RFs before and after reverberation removal, respectively. The filtered RFs were subsequently corrected for the effects of dipping and crustal anisotropy as described below.

Dipping interfaces and crustal anisotropy introduce systematic backazimuthal variations in the arrival times of the P_s phase and its crustal multiples, which can bias conventional H - κ estimates. To account for these effects, PRFs were first moveout-corrected to a reference slowness of 0.06 s/km and stacked in 5° backazimuth bins. Based on these stacked RFs, a five-dimensional grid search was performed for each phase to determine the optimal correction parameters associated with crustal anisotropy and dipping structures. Specifically, backazimuth-dependent arrival-time variations were parameterized using a $\cos 2\theta$ term associated with azimuthal anisotropy and a $\cos \theta$ term associated with a dipping interface and/or the plunge of the fast symmetric axis of anisotropy.

Backazimuthal variations in the phase arrival times were described as

$$F(\theta) = A_0 + A_1 \cos(\theta - \theta_1) - A_2 \cos 2(\theta - \theta_2) \quad (2)$$

where A_0 is the central arrival time, A_1 and θ_1 represent the amplitude and azimuth of the $\cos \theta$ term, and A_2 and θ_2 represent the amplitude and azimuth of the $\cos 2\theta$ term. These five parameters (A_0 , A_1 , A_2 , θ_1 , and θ_2) were determined through a five-dimensional grid search that maximized the coherence of the P_s phase and its crustal multiples (Eq. 2).

$$SUM_i(dt, A_1, A_2, \theta_1, \theta_2) = \left| \frac{1}{N} \sum_{\theta} [R'_G(t_i) \cdot |R'_G(t_i)|^{-\frac{3}{4}}] \right|^3 \cdot \frac{1}{N} \sum_{\theta} [R'_G(t_i) \cdot |R'_G(t_i)|^{-\frac{3}{4}}] \quad (3)$$

where i represents the phase (P_s , $PpPs$, and $PpSs+PsPs$), N is the number of stacked backazimuthal bins, and R'_G represents the moveout-corrected RFs stacked in each bin. The parameter $dt = A_0 - t_{\text{ref}}$ is the arrival-time correction relative to the theoretical arrival time (t_{ref}) predicted from conventional H - κ stacking.

The resulting correction parameters were used to remove the effects of dipping interfaces and crustal anisotropy from the phase arrivals prior to H - κ stacking. For stations without significant sedimentary reverberations, conventional H - κ stacking was applied to the corrected RFs to obtain the final estimates of H and κ (Figure S3). For stations affected by sedimentary reverberations, the time-corrected H - κ stacking formulation of Yu et al. (2015) was applied instead. Because the RFs used for the final stacking were stacked within

backazimuth bins after the dipping-interface and anisotropy corrections (R'_G), the sediment-induced P-arrival delayed-time (δt) and reverberation two-way traveltime (Δt) measured from individual RFs were averaged within each corresponding backazimuthal bin and the bin-averaged values were used in the corrected H - κ stacking. The bin-averaged values were then incorporated into the corrected H - κ stacking.

$$S(H, \kappa) = \sum_{\theta}^N w_1 R'_G(t_1 + \delta t_m) + w_2 R'_G(t_2 + \Delta t_m - \delta t_m) - w_3 R'_G(t_3 + \Delta t_m) \quad (4)$$

where t_1 , t_2 , and t_3 are arrival times of Ps , $PpPs$, and $PsPs+PpSs$, respectively, and δt_m and Δt_m are the bin-averaged values of δt and Δt , respectively, for the backazimuth θ . The crustal thickness (H) and V_p/V_s (κ) were determined by maximizing the $S(H, \kappa)$. For stations affected by sedimentary reverberations, the resulting H and κ represent the thickness and V_p/V_s of the crust beneath the sedimentary layer, respectively. This procedure assumes that the shallow sedimentary layer beneath each station is approximately flat and homogeneous (Figure S4).

Figure S3e illustrates the effect of the dipping interface and crustal anisotropy corrections at station JRB. In the uncorrected stacked RF (top), the Ps phase arrives at approximately 2.7 s, which is relatively early considering the crustal thickness of >33 km observed at surrounding stations (Figure 2b). In addition, the Moho multiple phases used in the H - κ stacking are poorly resolved. After applying the corrections (bottom), the Ps arrival shifts to approximately 3.2 s, reducing the anomalous early arrival, while the Moho multiple phases become more clearly identifiable. Consistent with the improved identification of the Moho multiple phases, the best-fitting κ increased from 1.437 to 1.526, partially reducing the magnitude of the anomalously low estimate. These changes improve the identification of the crustal phase arrivals used in the corrected H - κ stacking.

Figure S4h further illustrates the progressive effects of the corrections at a station affected by sedimentary reverberations, dipping interfaces, and crustal anisotropy. In the original stacked RF (top), delayed arrivals following the direct P phase indicate a pronounced influence of the sedimentary layer, and neither Ps nor its multiples from the Moho can be clearly identified. After the sedimentary reverberation correction (middle), the strong negative phase following the direct P arrival is substantially suppressed (Figure S4b), allowing the larger-amplitude arrival among the two previously merged positive peaks to be more clearly identified as the Pms phase. Following the additional corrections for the dipping interface and crustal anisotropy (bottom), the Ps and multiples from the Moho become more clearly resolved as individual arrivals. The theoretical arrival times t_1 , t_2 , and t_3 are also consistent with the corresponding observed phase arrivals. In addition, applying all corrections reduces uncertainty in the estimated H and κ (Figure S4e-f).

Text S3. Detailed parameters for alphaMELTS modeling: initial compositions and transition schemes

This section provides details on the initial mantle compositions and lithosphere–asthenosphere transition schemes used for the melt fraction estimation with the thermodynamic model pHMELTS (Asimow et al., 2004; Ghiorso et al., 2002), implemented in the alphaMELTS v.1.9 package (Smith & Asimow, 2005). The major element compositions of the asthenosphere were taken from the DMM values from Workman and Hart (2005). The model incorporated Si, Al, Fe, Mg, Ca, Na, Cr, Ti, and P, whereas Ni, Mn, and K were excluded to prevent calculation errors, as pHMELTS is not calibrated for NiO and MnO and often cannot accommodate K₂O in the absence of feldspar. In contrast, for the lithosphere, K was also included as a major element because it does not cause any calculation errors, given the presence of plagioclase during the lithospheric modeling. The lithospheric mantle composition was based on that of sample BR18, reconstructed using modal proportions and mineral chemistries of olivine, orthopyroxene, clinopyroxene, and spinel (Choi & Kwon, 2005).

To evaluate the sensitivity of melt fraction estimates to lithospheric composition, several other fertile samples (BR17, Choi & Kwon, 2005; BR4, Choi et al., 2005; SG14, Choi et al., 2002) were also tested in addition to sample BR18, which is a spinel lherzolite recovered from Baengnyeong Island (Choi & Kwon, 2005). The results are largely comparable to those utilizing BR18, except for slightly lower melt fractions at shallower depths for the SG14 model (Figure S7).

While the water content in the asthenosphere was taken from Workman and Hart (2005) along with the major elements, the water content in the lithosphere was independently estimated in this study due to the lack of previously reported water contents in Korean mantle xenoliths. The estimation was constrained to match the water contents in Cenozoic Korean magmas (1.5–2.0 wt.%; Park et al., 2022). The water contents in the magmas were calculated assuming the batch melting of the asthenosphere and lithosphere, followed by simple mixing of both melts. In addition, water was assumed to exhibit perfectly incompatible behavior. This calculation utilized the previously suggested melt fractions (lithosphere, 2–3%; asthenosphere, 0.5–1%) and mixing ratios (1:9–3:7) to reproduce the trace element ratios (e.g., La/Yb, Nb/La) of Cenozoic Korean volcanic rocks (Lee et al., 2025). Accordingly, the H₂O content in the lithosphere was assumed to be 500 ppm, yielding 1.16–2.29 wt.% H₂O in the melt, which is consistent with the water contents in Cenozoic Korean magmas.

As described in the main text, the bulk melt fraction at each step was calculated assuming a linear transition between the asthenosphere and lithosphere (Figure 5). However, because the actual ratio of asthenosphere to lithosphere within the transition zone is poorly constrained, exponential and reverse-exponential changes were also tested here. The results exhibit only a slight difference in maximum melt fractions due to discrepancies in the proportions of hydrous lithosphere at the LAB (Figure S8).

Text S4. Assessment of consistency between PRF- and SRF-derived Moho depth estimates
To assess the consistency between Moho depth estimates obtained from the two independent methods (H - κ stacking of PRFs and depth conversion of SRFs), we compared their differences relative to their combined uncertainties (Figures 2a and 2b). Assuming that the uncertainties associated with the two methods are independent, the uncertainty of the difference was estimated by standard error propagation as

$$\sigma_{\Delta} = \sqrt{\sigma_1^2 + \sigma_2^2} \quad (5)$$

where σ_1 and σ_2 represent the uncertainties associated with the PRF- and SRF-derived Moho depth estimates, respectively. Although these uncertainties were derived using different approaches (curvature analysis for H - κ stacking and bootstrap resampling for SRF depth conversion), both values were regarded as representative metrics for the uncertainty of their respective models.

We then performed a Z-test and defined a Z-score as:

$$Z = \frac{|H_1 - H_2|}{\sigma_{\Delta}} \quad (6)$$

where H_1 and H_2 denote the Moho depths derived from each method. The results of the Z-test demonstrate that the Moho depths derived from the two independent methods are statistically consistent. Specifically, all stations except GKP2 ($Z = 2.37$) satisfied the condition $Z < 1.96$, confirming that the two methodologies yield statistically consistent results within a 95% confidence interval. GKP2 is located in the western Gyeongsang Basin, where the regional Moho does not show pronounced lateral variation (Chang and Baag, 2007). Inspection of the individual RFs shows that the Moho-related phase in the SRFs exhibits greater arrival-time variability than in the PRFs, potentially reflecting station-specific uncertainty associated with variations in signal quality among individual SRFs. This variability may contribute to the discrepancy between the two Moho estimates. The high degree of correspondence is further illustrated in the spatial distribution and histogram of the normalized Z-scores ($Z/1.96$), as shown in Figure S9.

Text S5. Monte Carlo uncertainty propagation for regional differences in Moho depth, V_P/V_S , and LAB depth

To evaluate whether the observed regional differences in Moho depth, V_P/V_S , and LAB depth are robust relative to station-specific uncertainties, we performed Monte Carlo uncertainty propagation analyses (Figure S10). Station-level uncertainties for Moho depth and V_P/V_S were estimated from $H-k$ stacking, whereas LAB depth uncertainties were estimated as the standard deviation of depths obtained from 1,000 bootstrap resamplings, each using 80% of traces randomly resampled with replacement. For each station, each parameter was randomly sampled from a normal distribution centered on the estimated value, with a standard deviation given by the corresponding station-level uncertainty. Stations along the eastern coast (KOSB, KSA, GEJB, YAYB, OKEB, SNGB, IMWB, ULJ2, YODB, JEJB, SKC, YAYA, JMJ, KAWA, TOHA, UJN, GGGA, PHA2, and HMGA) were excluded from this analysis because they exhibit exceptionally shallow Moho depth (~25 km) and high crustal V_P/V_S ratios, compared with the other regions (Figures 2b and 2c). In each realization, the mean values for the two regions were calculated, and their differences were evaluated. This procedure was repeated 10,000 times to obtain the probability distribution of the mean difference for each parameter.

The Moho depth difference, defined as (YM + GB) minus (GM + H-IB), has a mean value of 1.7 km and a 95% confidence interval of 0.9–2.5 km. The probability that the Moho beneath the YM and GB is deeper than that beneath the GM and H-IB is 0.9999, indicating that the observed Moho depth contrast between the two regions appears robust to station-level uncertainties.

Crustal V_P/V_S also shows a systematic regional contrast. The difference has a mean value of 0.04 and a 95% confidence interval of 0.02–0.05. The probability that the difference is greater than zero exceeds 0.9999, indicating that V_P/V_S is consistently higher in the YM and GB region than in the GM and H-IB region within the estimated uncertainties.

The LAB depth shows the opposite depth pattern to the Moho. The mean sampled depths are 71.5 km beneath the GM and H-IB and 68.0 km beneath the YM and GB, with 95% confidence intervals of 70.4–72.5 km and 67.0–69.1 km, respectively. The difference is 3.4 km, with a 95% confidence interval of 2.0–4.9 km, indicating that the LAB beneath the YM and GB is consistently shallower than that beneath the GM and H-IB. The probability that the LAB beneath the YM and GB is shallower than that beneath the GM and H-IB exceeds 0.9999, indicating that the observed LAB depth contrast is robust to station-level uncertainties.

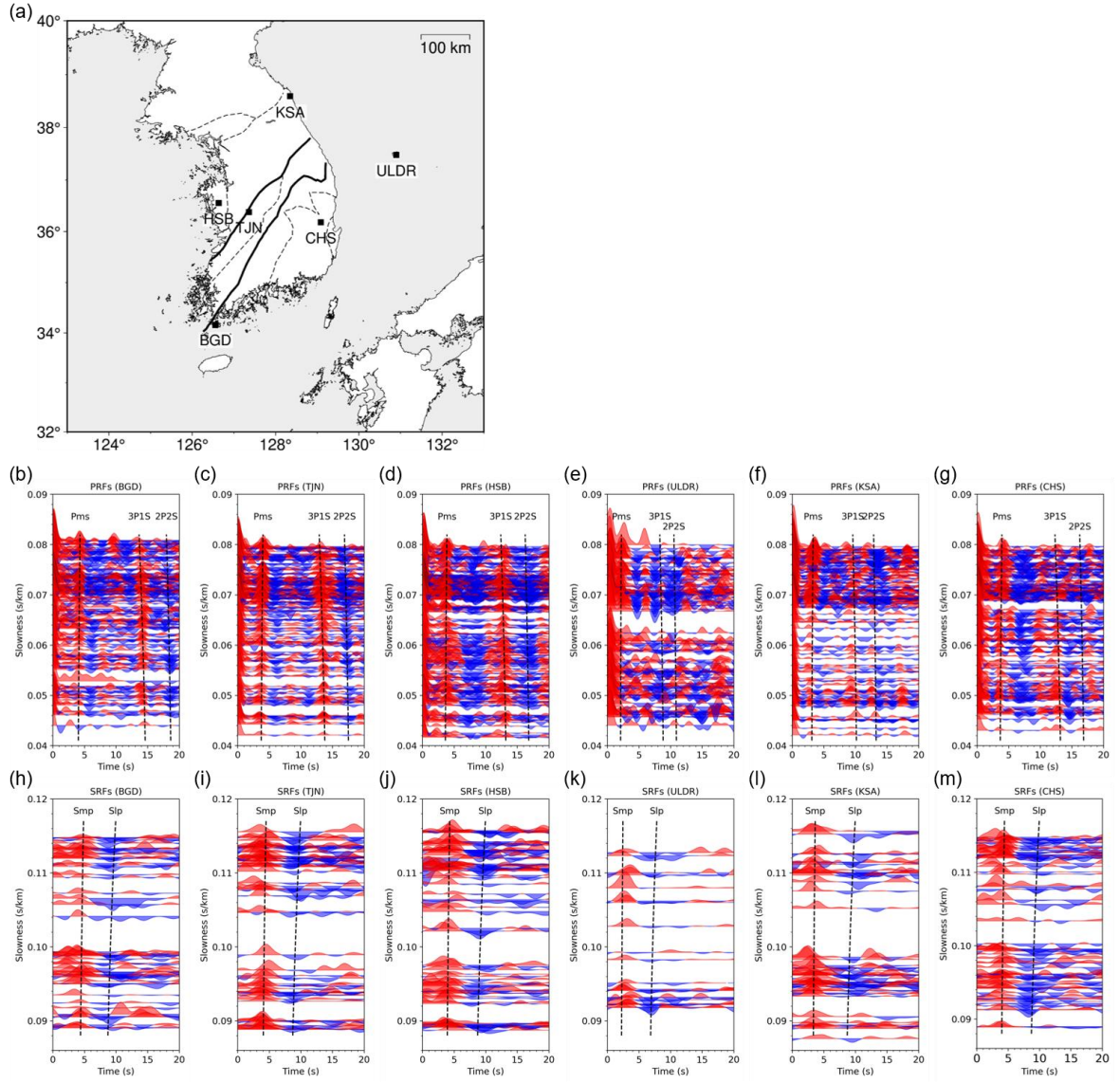


Figure S1. Representative individual PRFs and SRFs after quality control. (a) Map showing the locations of the seismic stations (BGD, HSB, TJN, ULDR, KSA, and CHS) used for (b)–(m). (b)–(g) Individual PRFs, with dashed lines indicating theoretical arrival times of the P -to- S converted phase from the Moho (Pms) and crustal reverberation phases ($3P1S$ ($PpPs$) and $2P2S$ ($PpSs+PsPs$)) between the Moho and the surface. (h)–(m) Individual SRFs, with dashed lines indicating theoretical arrival times of the S -to- P converted phases from the Moho (Smp) and the LAB (Slp).

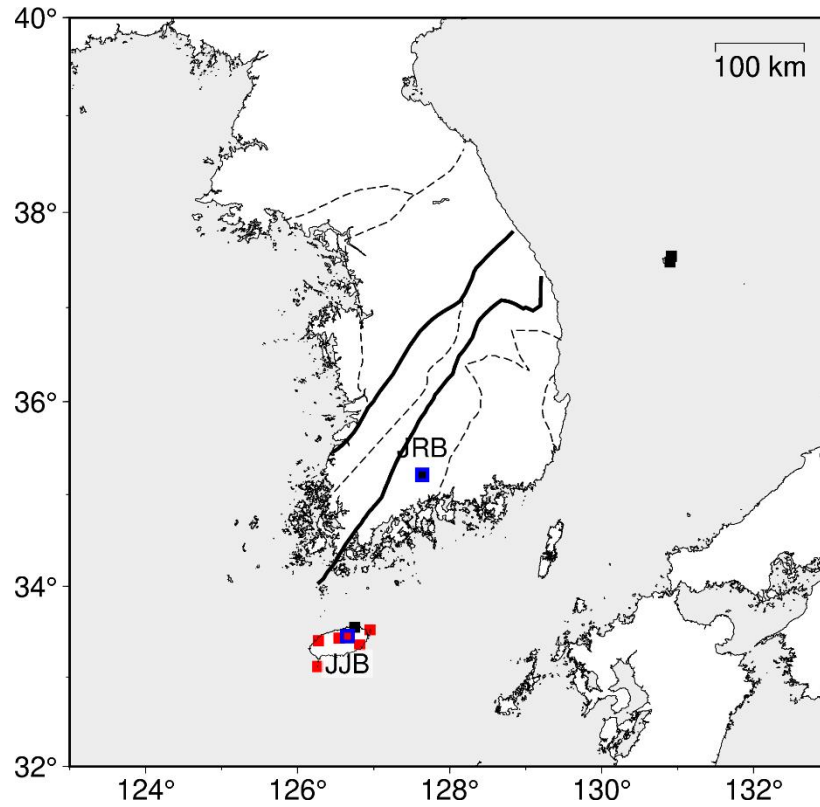


Figure S2. Map showing locations of seismic stations analyzed using modified H - κ stacking methods of PRFs for estimating crustal thickness and V_P/V_S . Grey squares denote stations analyzed using anisotropy- and dip-corrected H - κ - c stacking after sediment correction (Yu et al., 2015; Li et al., 2019). Three red squares denote stations analyzed using the H - κ - c stacking (Li et al., 2019). Blue squares indicate stations JRB and JJB, whose stacking results are shown in Figures S3 and S4. JRB is a broadband station located on Mount Jiri.

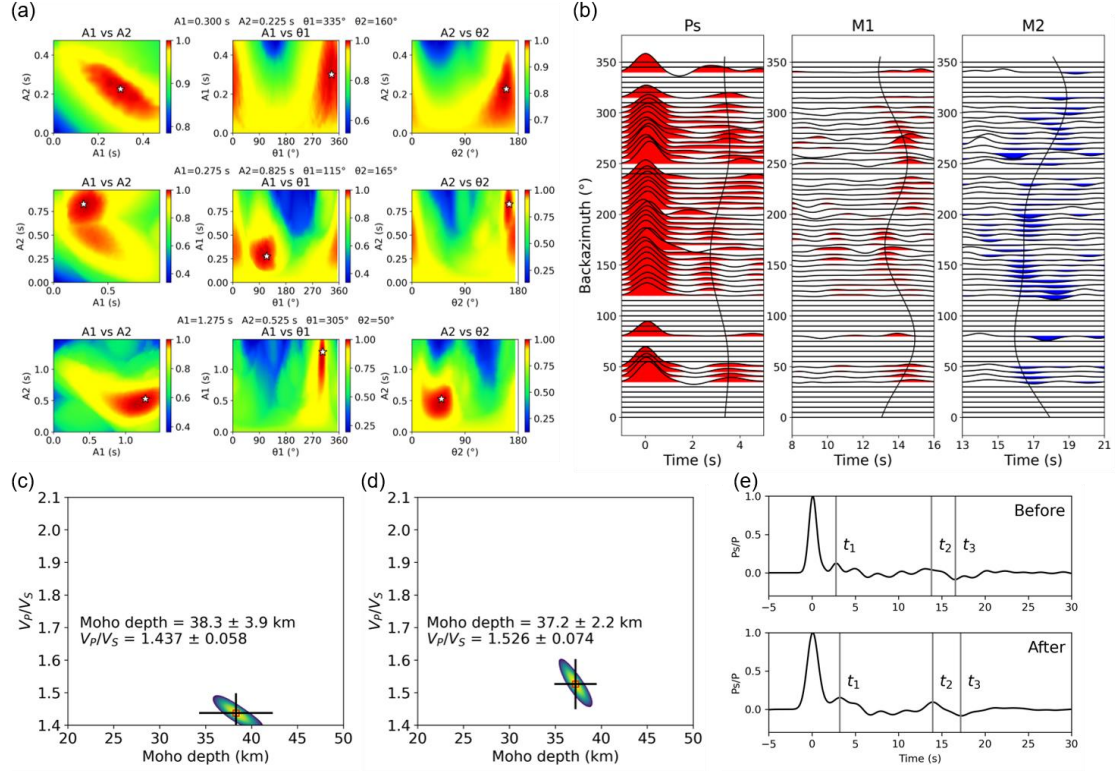


Figure S3. H – κ – c stacking result for station JRB following Li et al. (2019). (a) Energy maps obtained from a grid search of the curve-fitting parameters. (b) Curve-fitting results for P_s , 3P1S (M1; P_pP_s) and 2P2S (M2; $P_pS_s+P_sP_s$) phases from the Moho. (c) Conventional H – κ stacking result without correction for dipping interface and seismic anisotropy, yielding a crustal thickness of 38.3 ± 3.9 km and a V_P/V_S of 1.437 ± 0.058 . (d) H – κ – c stacking result, indicating a crustal thickness of 37.2 ± 2.2 km and a V_P/V_S of 1.526 ± 0.074 . (e) Stacked PRF for station JRB before (top) and after (bottom) corrections for the dipping interface and seismic anisotropy, with gray lines indicating the theoretical arrival times of the P_s (t_1), 3P1S (t_2), and 2P2S (t_3) phases from the Moho, calculated from the results shown in (c) and (d), respectively.

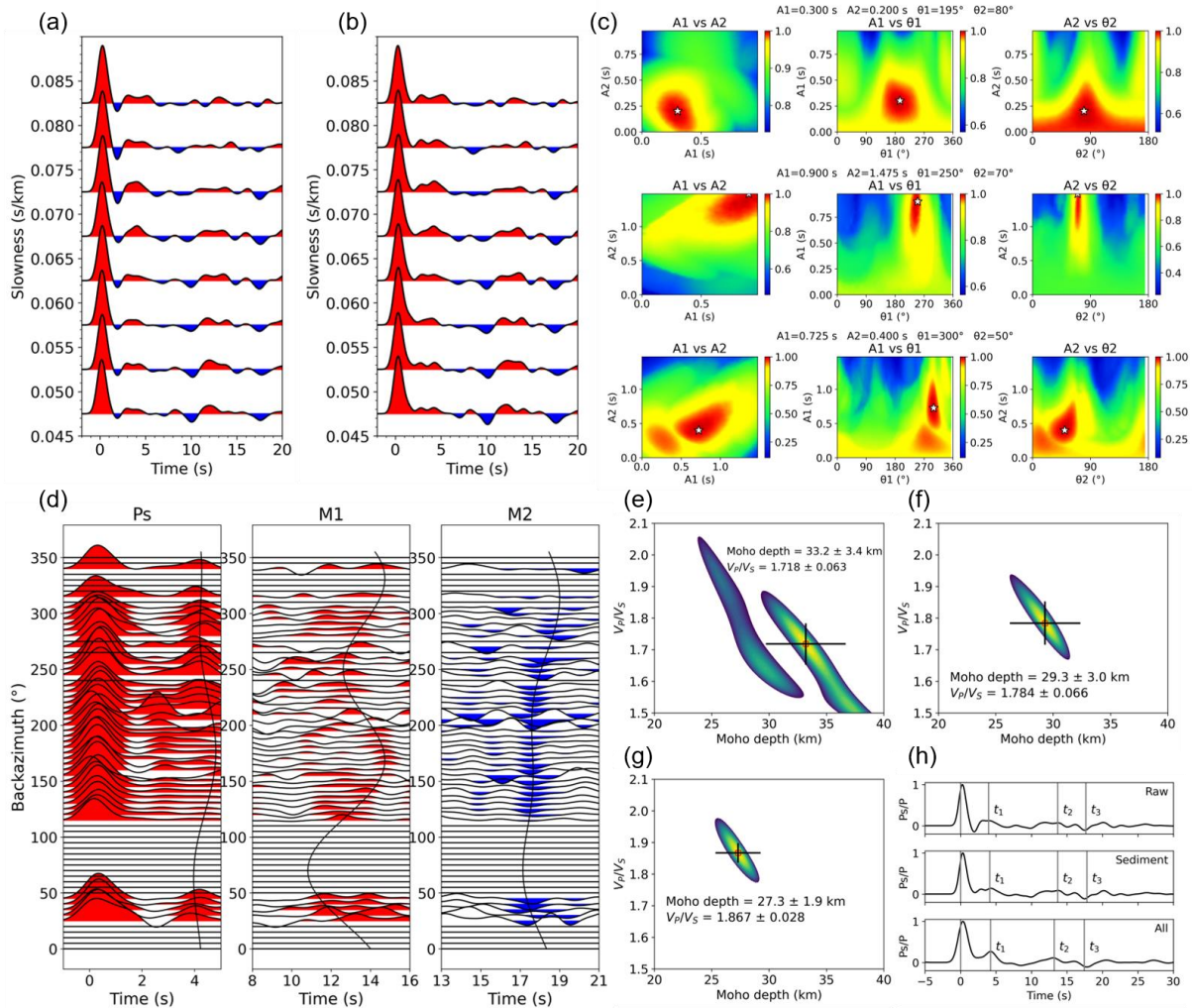


Figure S4. H - κ - c stacking results with sediment correction for station JJB. (a) Raw PRFs stacked in slowness bins of 0.005 s/km. (b) PRFs after correction for reverberation phases generated within the sedimentary layer, following Yu et al. (2015). (c) Energy maps obtained from a grid search of the curve-fitting parameters. (d) Curve fitting results for P_s , 3P1S (M1; $PpPs$) and 2P2S (M2; $PpSs+PsPs$) phases from Moho. (e) Conventional H - κ stacking result (Zhu & Kanamori, 2000) with raw PRFs shown in (a), indicating a Moho depth of 33.2 ± 3.4 km and a V_P/V_S of 1.718 ± 0.063 . (f) Sediment-corrected H - κ stacking result (Yu et al., 2015) without correction for dipping interface and seismic anisotropy, yielding a crustal thickness of 29.3 ± 3.0 km and a V_P/V_S of 1.784 ± 0.066 . (g) H - κ - c stacking with correction for dipping interfaces and seismic anisotropy (Li et al., 2019) after applying sediment correction (Yu et al., 2015), yielding a crustal thickness of 27.3 ± 1.9 km and a V_P/V_S of 1.867 ± 0.028 . (h) Stacked PRFs before correction (top), after correction for the sedimentary layer (middle), and after applying corrections for the sedimentary layer, dipping interfaces, and anisotropy (bottom). Gray lines indicate the theoretical arrival times of the P_s (t_1), 3P1S (t_2), and 2P2S (t_3) phases, calculated using the corresponding H and V_P/V_S values obtained from the stacking results in (e), (f), and (g), respectively.

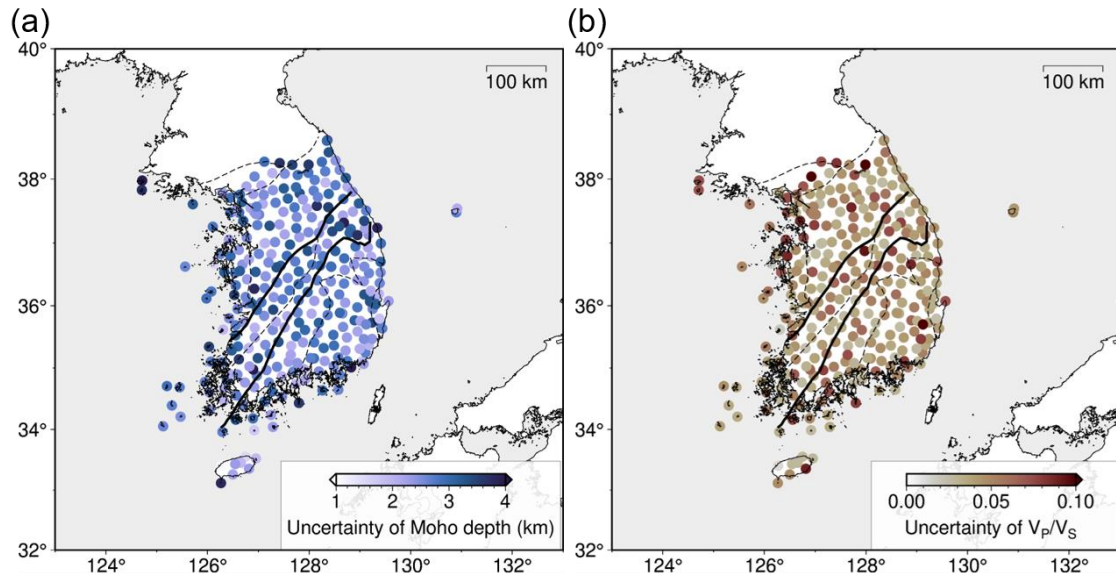


Figure S5. Uncertainties of H - k stacking analysis of PRFs, estimated from the curvature of the stacking function (Zhu & Kanamori, 2000). (a) Uncertainty of Moho depth. (b) Uncertainty of crustal average V_P/V_S .

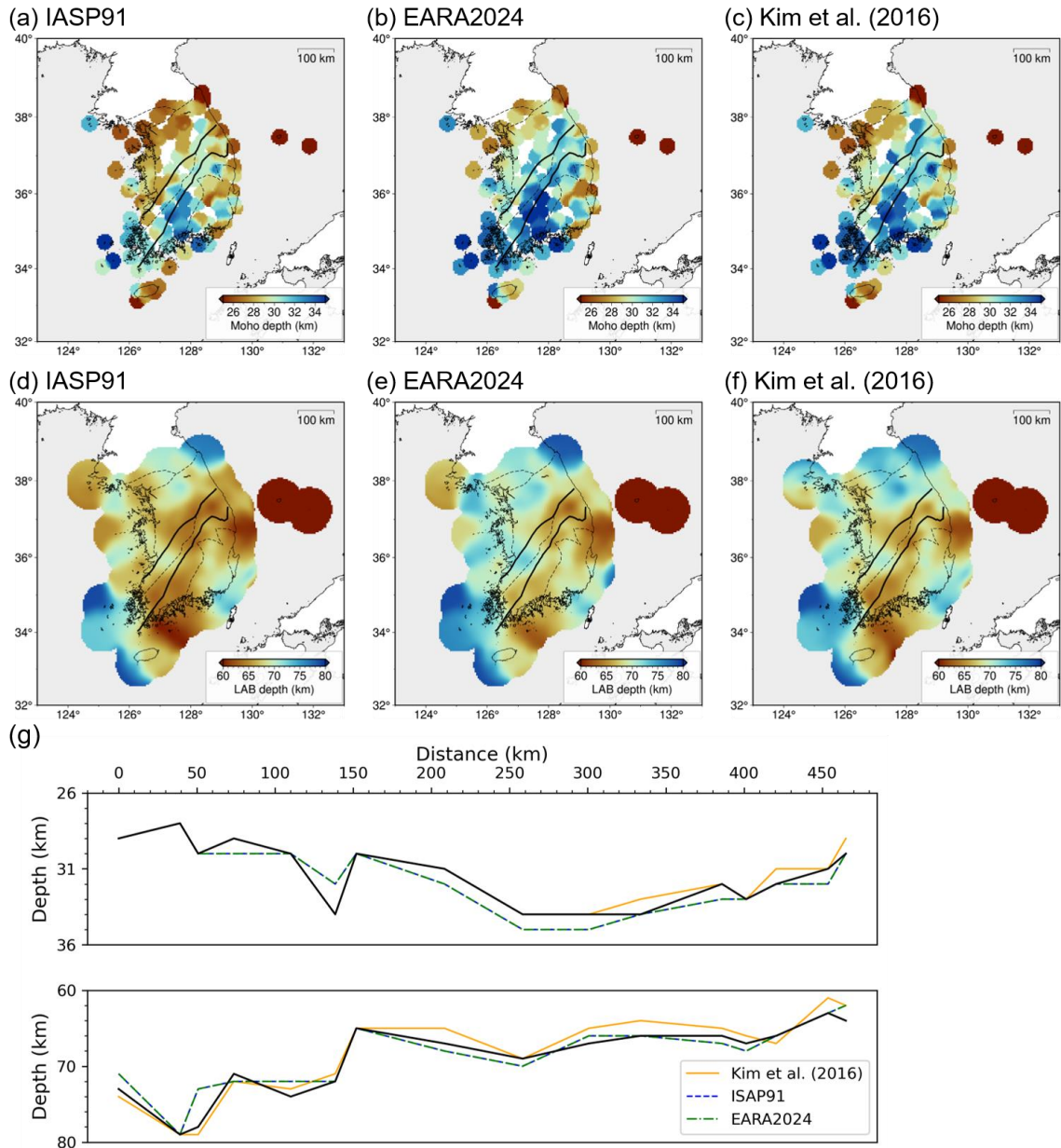


Figure S6. Moho and LAB depths estimated from depth-converted SRFs using different velocity models. (a–c) Moho depths from (a) IASP91 (Kennett & Engdahl, 1991), (b) EARA2024 (Xi et al., 2024) based on full-waveform inversion tomography, and (c) the ambient-noise tomography model (Kim et al., 2016). (d–f) LAB depths from (d) IASP91, (e) EARA2024, and (f) the ambient-noise tomography model. (g) Moho (top) and LAB (bottom) depths along the P–P' profile in Figure 3a. The black line denotes results from the hybrid model combining the tomography model of Kim et al. (2016) and EARA2024 (Xi et al., 2024) as described in Section 2.3, while the orange solid, blue dashed, and green dash-dotted lines show the depth estimates obtained using Kim et al. (2016), IASP91, and EARA2024, respectively.

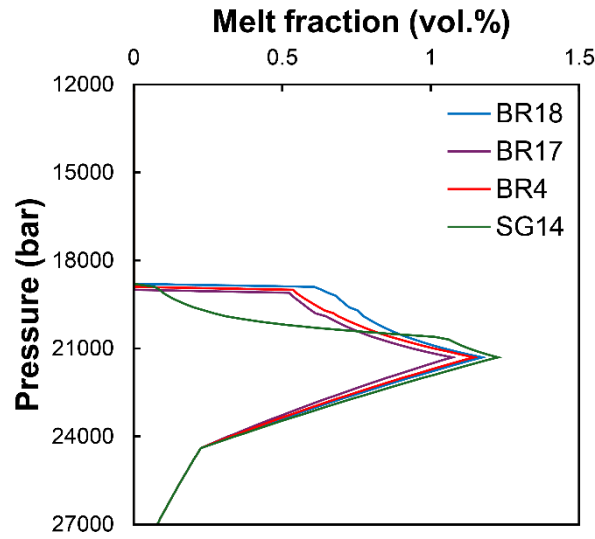


Figure S7. Thermodynamic modeling results showing the sensitivity of melt fraction estimates to lithospheric composition. The compositions of four different fertile mantle xenolith samples (BR18 and BR17, Choi & Kwon, 2005; BR4, Choi et al., 2005; SG14, Choi et al., 2002) are utilized. The pressure-temperature conditions are the same as those used for modeling the NW block in Figure 5a. For clarity, only results for a single potential temperature of 1300°C are shown.

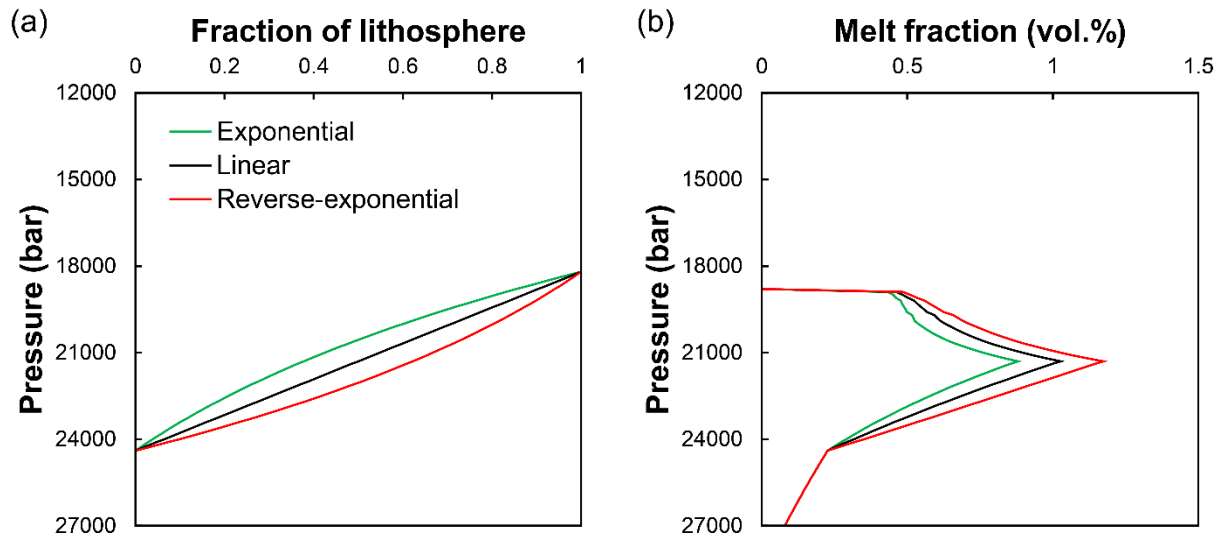


Figure S8. Sensitivity of melt fraction estimates from thermodynamic modeling to different lithosphere-asthenosphere transition models. (a) Lithospheric fraction profiles as a function of pressure for three mixing models (exponential, linear, and reverse-exponential), assuming a transition thickness of 20 km. (b) Bulk melt fraction profiles estimated from the three mixing models shown in (a). The pressure-temperature conditions are the same as those used for modeling the NW block in Figure 5a. For clarity, only results for a single potential temperature of 1300°C are shown.

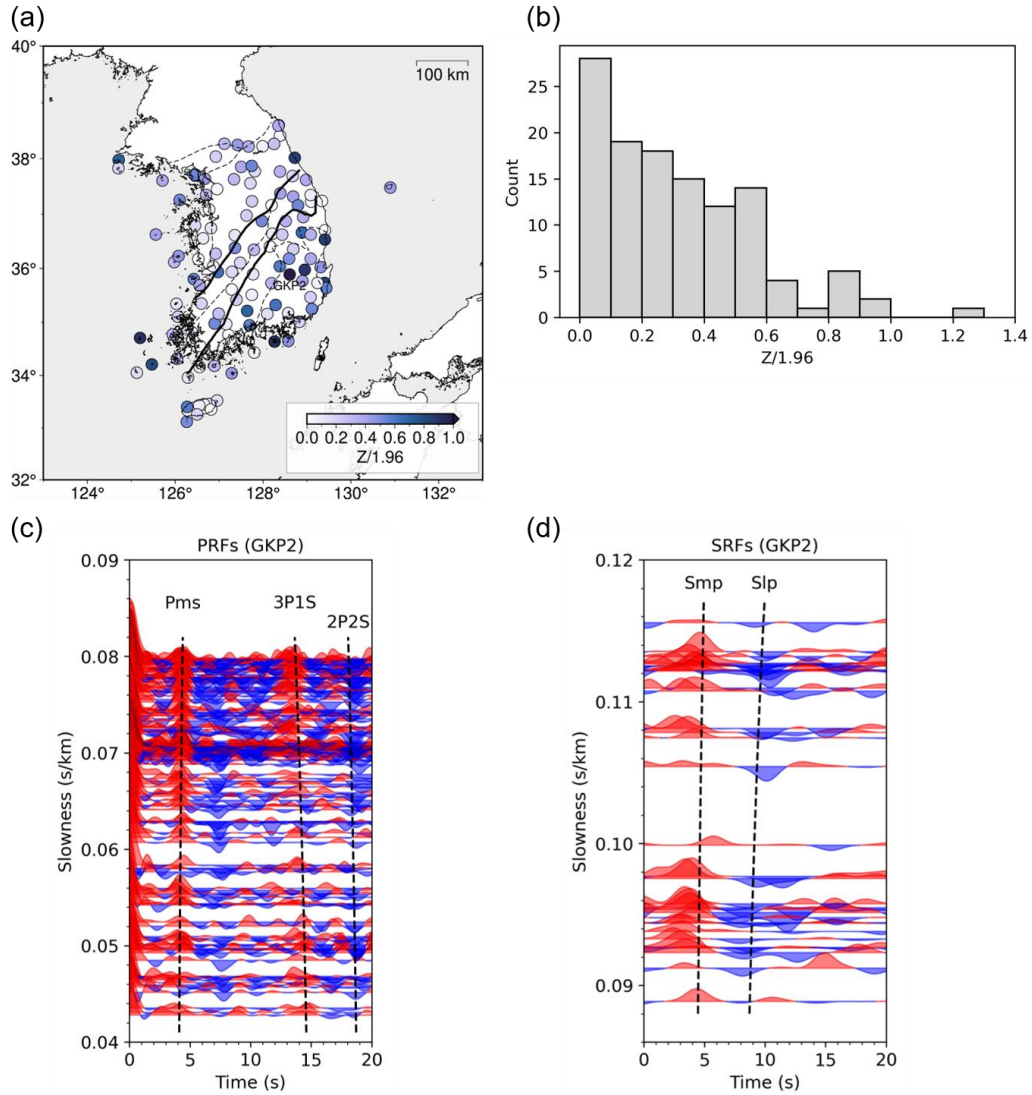


Figure S9. Comparison between Moho depths derived from the H - κ stacking of PRFs and depth-converted SRFs. (a) Spatial distribution of the consistency between the two methods. The map shows the Z-scores normalized by the 95% confidence threshold (1.96) for each seismic station. Only one station, GKP2, exhibits a normalized Z-score ($Z/1.96$) exceeding 1.0. (b) Histogram of the normalized Z-scores. Values of $Z/1.96 < 1.0$ indicate that the Moho depths estimated from the two independent methods are statistically consistent within the 95% confidence interval. Individual (c) PRFs and (d) SRFs at station GKP2.

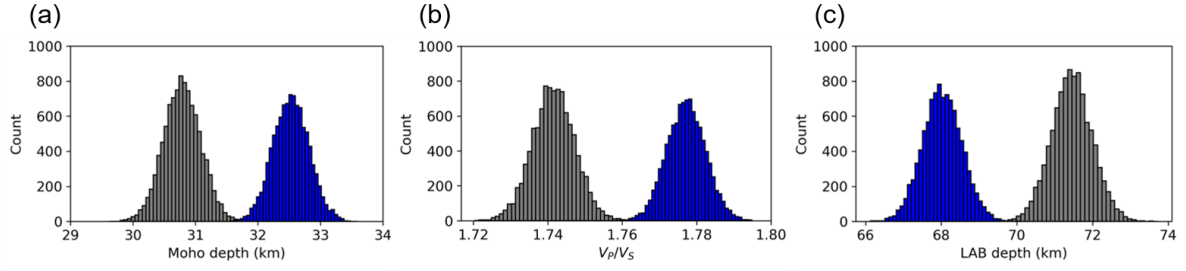


Figure S10. Monte Carlo uncertainty propagation for regional differences in (a) Moho depth, (b) crustal V_P/V_S , and (c) LAB depth. Histograms show distributions from 10,000 Monte Carlo realizations of the regional mean values of each parameter. Gray and blue bars represent the NW and SE blocks, respectively. Station-level uncertainties were incorporated by randomly sampling each station value from a normal distribution defined by its estimated value and corresponding uncertainty.

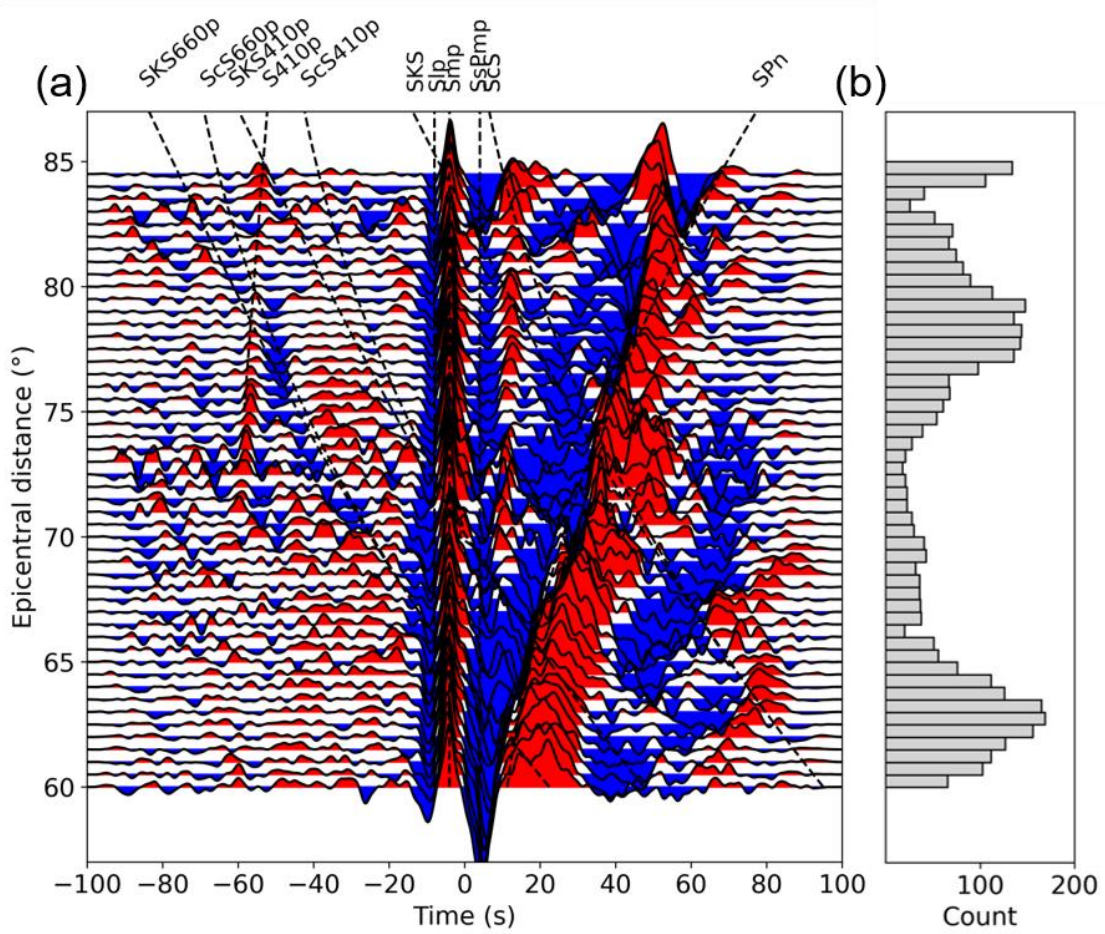


Figure S11. Stacked SRFs as a function of epicentral distance. (a) SRFs from all stations used in this study, stacked in 0.5° epicentral-distance bins. The *S*-to-*P* converted phases from the Moho (*Smp*) and LAB (*S/p*) are the primary phases used in this study. Other labeled phases include *S410p*, *SKS410p*, *ScS410p*, *SKS660p*, *ScS660p*, *SKS*, *SsPmp*, *ScS*, and *SPn*. *SPn* corresponds to an *S*-to-*P* conversion propagating horizontally through the uppermost mantle beneath the receiver side, whereas *SsPmp* exhibits relatively large amplitudes due to its post-critical reflection characteristic (e.g. Liu et al., 2018). (b) Histogram of the number of SRFs in each epicentral-distance bin.

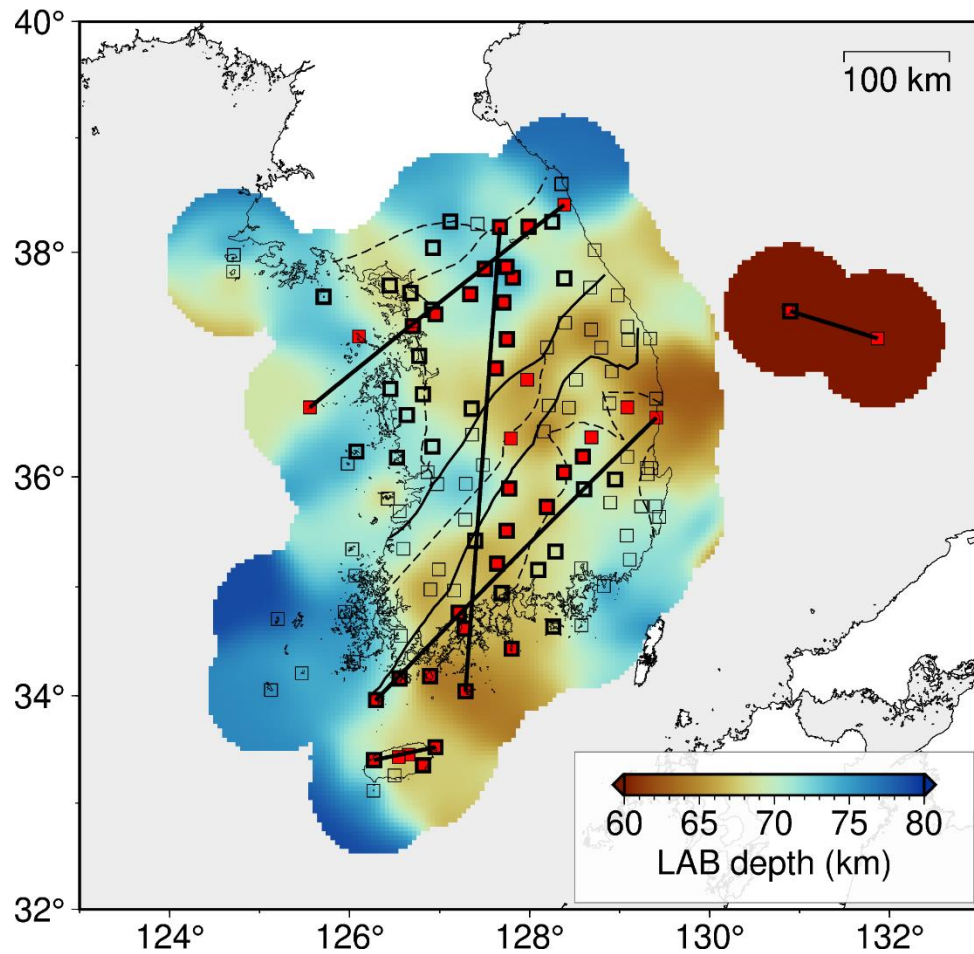


Figure S12. Seismic stations used in the SRF analyses. Red squares mark stations used to construct the SRF profiles shown in Figure 3. Squares with thick outlines indicate stations used for SRF modeling and for generating the stacked SRFs shown in Figure 4.

References

- Asimow, P. D., J. E. Dixon, & C. H. Langmuir (2004), A hydrous melting and fractionation model for mid-ocean ridge basalts: Application to the Mid-Atlantic Ridge near the Azores, *Geochemistry, Geophysics, Geosystems*, 5, <https://doi.org/10.1029/2003GC000568>.
- Chang, S.-J., & C.-E. Baag (2007), Moho depth and crustal Vp/Vs variation in southern Korea from teleseismic receiver functions: Implication for tectonic affinity between the Korean Peninsula and China, *Bulletin of the Seismological Society of America*, 97, 1621–1631, <https://doi.org/10.1785/0120050264>.
- Choi, S. H., & S.-T. Kwon (2005), Mineral chemistry of spinel peridotite xenoliths from Baengnyeong Island, South Korea, and its implications for the paleogeotherm of the uppermost mantle, *Island Arc*, 14, 236–253, <https://doi.org/10.1111/j.1440-1738.2005.00469.x>.
- Choi, S. H., J. I. Lee, C.-H. Park, & J. Moutte (2002), Geochemistry of peridotite xenoliths in alkali basalts from Jeju Island, Korea, *Island Arc*, 11, 221–235, <https://doi.org/10.1046/j.1440-1738.2002.00367.x>.
- Choi, S. H., S. T. Kwon, S. B. Mukasa, & H. Sagong (2005), Sr-Nd-Pb isotope and trace element systematics of mantle xenoliths from Late Cenozoic alkaline lavas, South Korea, *Chemical Geology*, 221, 40–64, <https://doi.org/10.1016/j.chemgeo.2005.04.008>.
- Ghiorso, M. S., M. M. Hirschmann, P. W. Reiners, & V. C. Kress III (2002), The pMELTS: A revision of MELTS for improved calculation of phase relations and major element partitioning related to partial melting of the mantle to 3 GPa, *Geochemistry, Geophysics, Geosystems*, 3, 1–35, <https://doi.org/10.1029/2001GC000217>.
- Kennett, B. L. N., & E. R. Engdahl (1991), Traveltimes for global earthquake location and phase identification, *Geophysical Journal International*, 105, 429–465, <https://doi.org/10.1111/j.1365-246X.1991.tb06724.x>.
- Kim, S., H. Tkalčić, J. Rhie, & Y. Chen (2016), Intraplate volcanism controlled by back-arc and continental structures in NE Asia inferred from transdimensional Bayesian ambient noise tomography, *Geophysical Research Letters*, 43, 8390–8398, <https://doi.org/10.1002/2016GL069483>.
- Lee, W., H. Lee, H. Kim, J.-H. Song, J. Hong, J. Park, H. Jung, J. Lee, N. Takahata, Y. Sano, & T. P. Fischer (2025), Water and gas geochemistry of springs in Ulleungdo volcano, South Korea: Implications for degassing of upper mantle-derived volatiles in Northeast Asia, *Journal of Hydrology*, 647, 132286, <https://doi.org/10.1016/j.jhydrol.2024.132286>.
- Li, J., X. Song, P. Wang, & L. Zhu (2019), A generalized H–k method with harmonic corrections on Ps and its crustal multiples in receiver functions, *Journal of Geophysical Research: Solid Earth*, 124, 3782–3801, <https://doi.org/10.1029/2018JB016356>.
- Liu, T., S. L. Klemperer, C. Yu, & J. Ning (2018), Post-critical SsPmp and its applications to Virtual Deep Seismic Sounding (VDSS)—1: Sensitivity to lithospheric 1-D and 2-D structure, *Geophysical Journal International*, 215, 880–894, <https://doi.org/10.1093/gji/ggy307>.

- Park, J., H. Lim, B. Myeong, Y.-D. Jang, & M. Brenna (2022), Basaltic cognate enclaves from Dokdo Island as a window for intraplate mafic alkaline OIB magma dynamics in a back-arc basin, *Contributions to Mineralogy and Petrology*, 177, <https://doi.org/10.1007/s00410-022-01951-4>.
- Smith, P. M., & P. D. Asimow (2005), Adiatat_1ph: A new public front-end to the MELTS, pMELTS, and pHMELTS models, *Geochemistry, Geophysics, Geosystems*, 6, <https://doi.org/10.1029/2004GC000816>.
- Workman, R. K., & S. R. Hart (2005), Major and trace element composition of the depleted MORB mantle (DMM), *Earth and Planetary Science Letters*, 231, 53–72, <https://doi.org/10.1016/j.epsl.2004.12.005>.
- Xi, Z., M. Chen, S. S. Wei, J. Li, T. Zhou, B. Wang, & Y. Kim (2024), EARA2024: A new radially anisotropic seismic velocity model for the crust and upper mantle beneath East Asia and Northwestern Pacific subduction zones, *Geophysical Journal International*, 239, 914–935, <https://doi.org/10.1093/gji/ggae302>.
- Yu, Y., J. Song, K. H. Liu, & S. S. Gao (2015), Determining crustal structure beneath seismic stations overlying a low-velocity sedimentary layer using receiver functions, *Journal of Geophysical Research: Solid Earth*, 120, 3208–3218, <https://doi.org/10.1002/2014JB011610>.
- Zhu, L., & H. Kanamori (2000), Moho depth variation in southern California from teleseismic receiver functions, *Journal of Geophysical Research: Solid Earth*, 105, 2969–2980, <https://doi.org/10.1029/1999JB900322>.